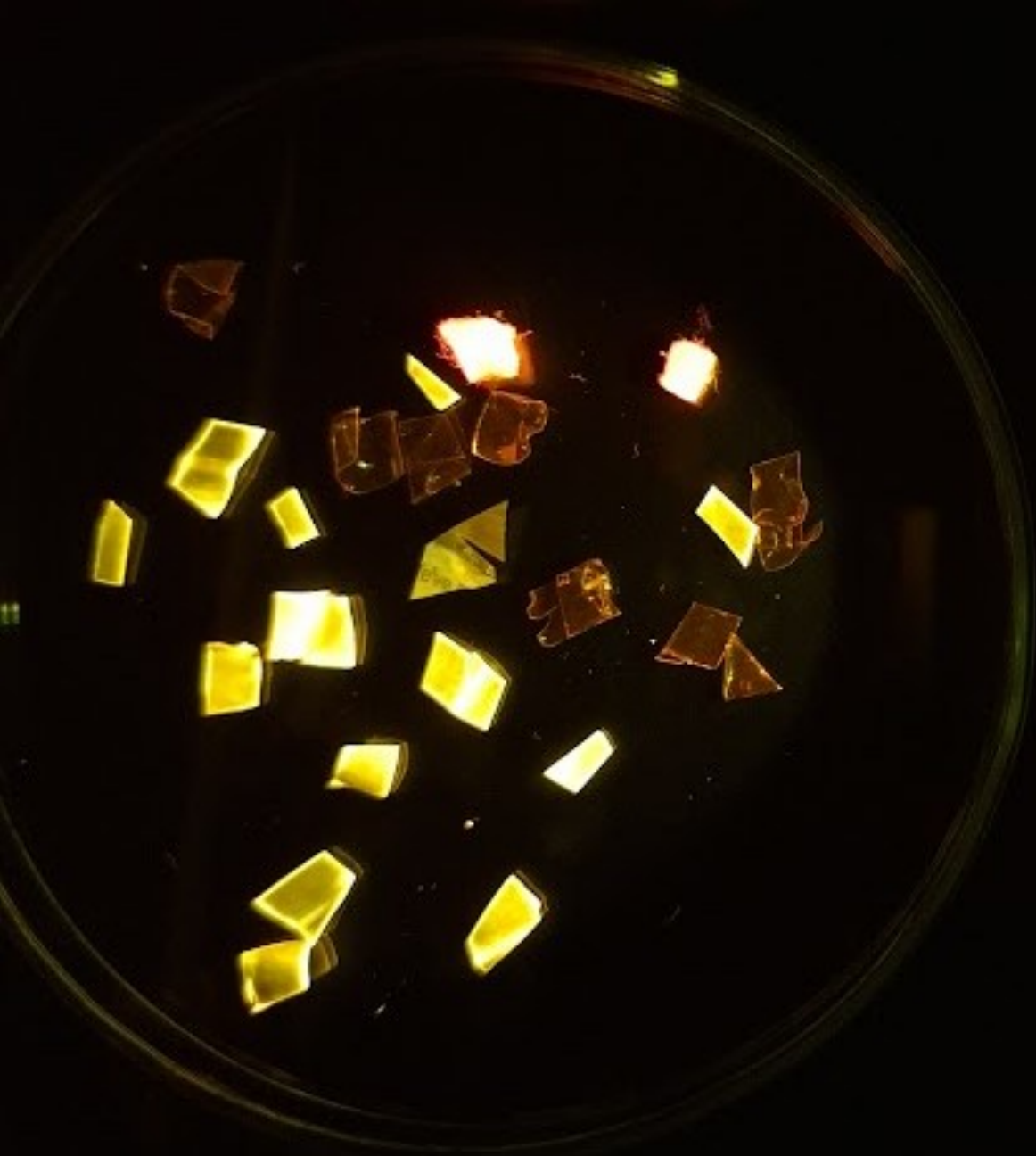




ARTIFICIAL INTELLIGENCE FOR ONSITE ANALYSIS OF MICROPLASTICS IN URBAN WATER RUNOFF

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CHEMISTRY WITH ARTIFICIAL INTELLIGENCE



INTRODUCTION

MICROPLASTICS... small plastic particles, synthetic polymers

CHALLENGES... time consuming process, in-lab based traditional spectroscopy methods

AIMS... to build a machine learning model for the rapid identification and quantification of microplastics using images obtained from a smartphone



DATA COLLECTION, PREPARATION, PROCESSING

ARTIFICIAL SAMPLING

Primary sourcing various types of plastics from household waste and organic materials from the wild.

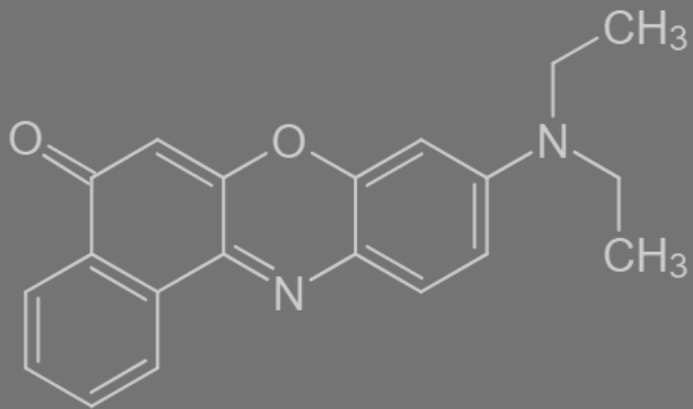


PET, Polyester, LDPE, HDPE, Grass, PS, PMMA, PP, cig filter, microbeads, PVC, plastic nets

DATA
COLLECTION,
PREPARATION,
PROCESSING

STAINING

Fluorescence staining of various microplastic samples using **NILE RED** and **ETHANOL** solution of concentration 1 mg/mL, for one hour in water medium.



DATA
COLLECTION,
PREPARATION,
PROCESSING

IMAGING

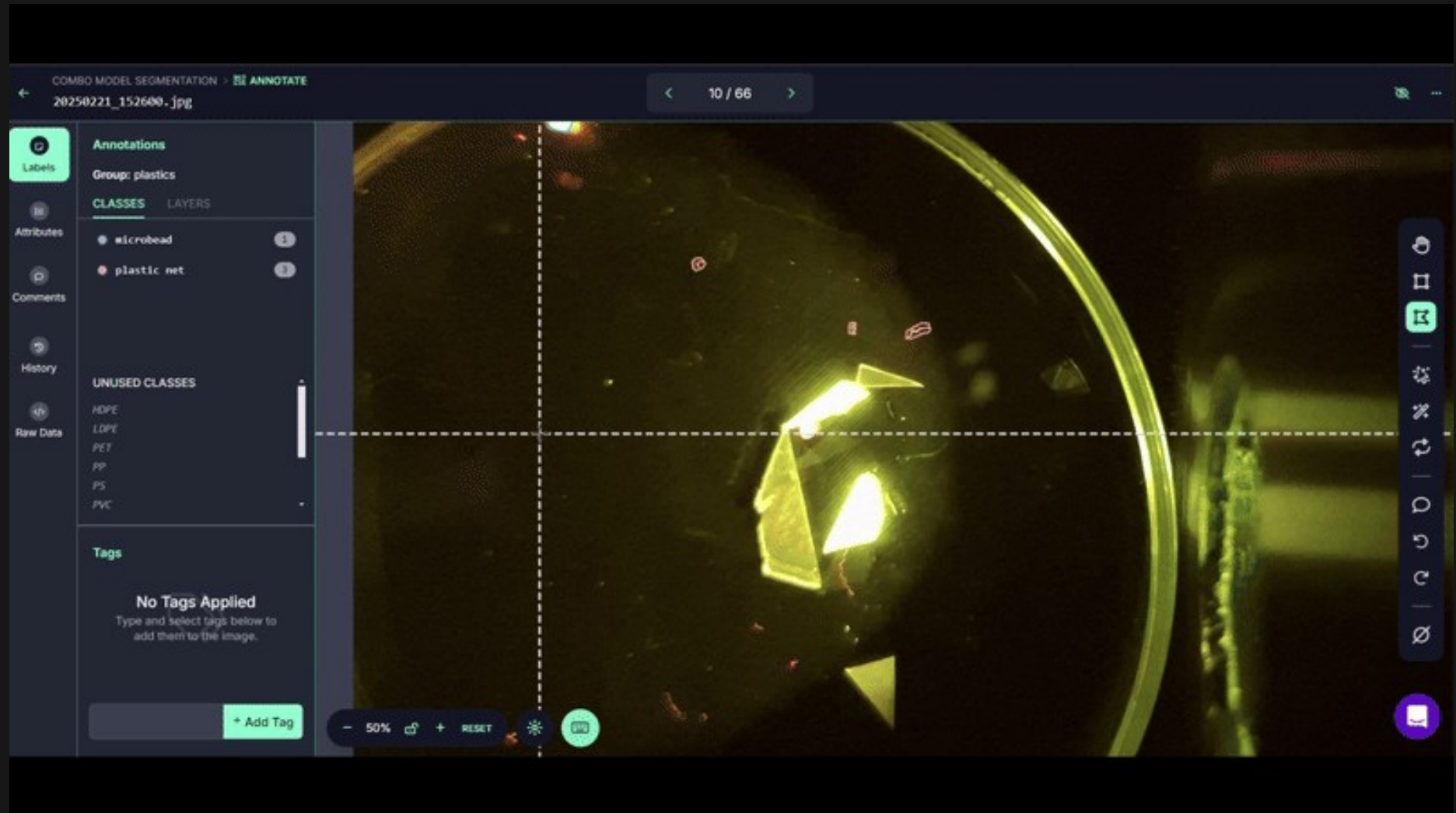
Under the stimulation of **BLUE LIGHT** at 450 nm and the use of an acrylic **ORANGE FILTER** in a 3D printed **BLACK BOX**. An image is captured using a smartphone.



DATA
COLLECTION,
PREPARATION,
PROCESSING



LABELLING

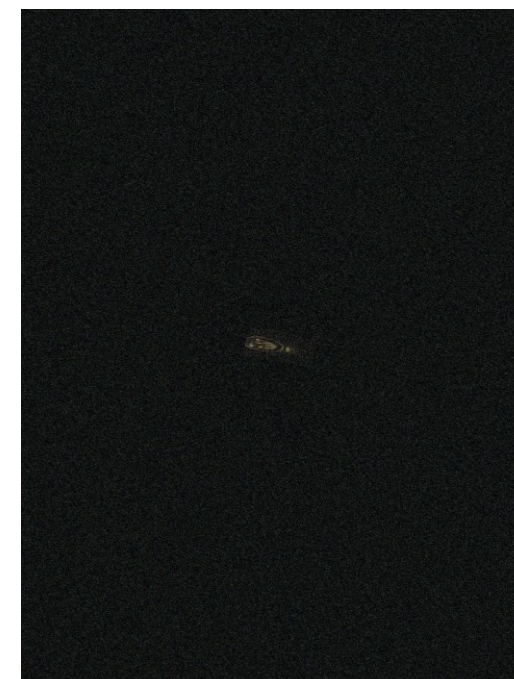
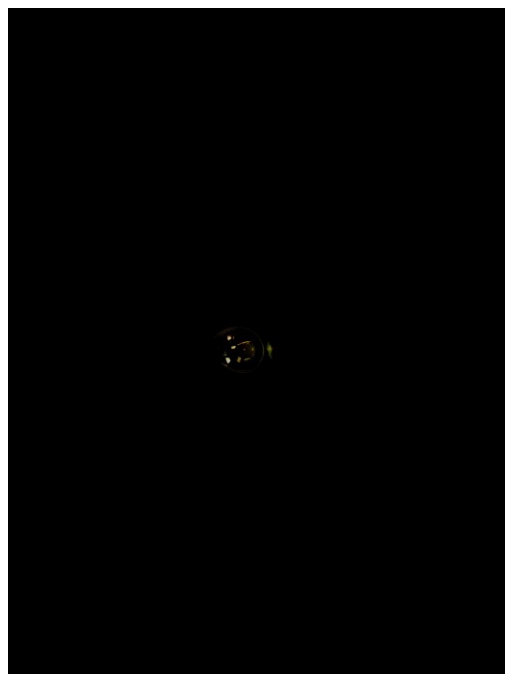
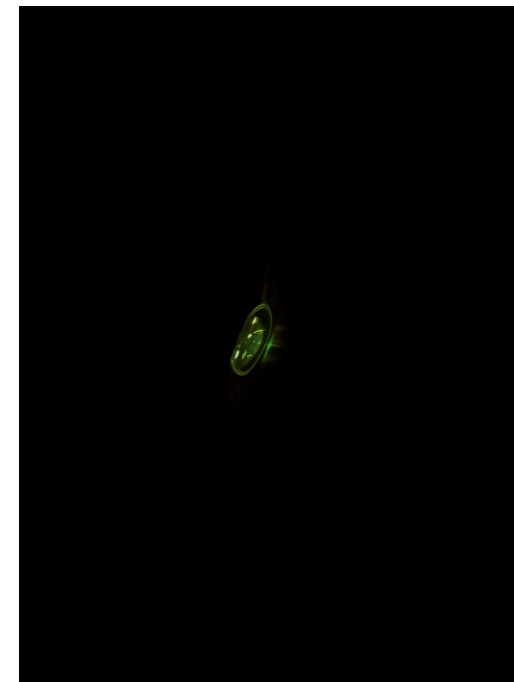


DATA AUGMENTATION

Only 405 labelled images...

The Python library
“Albumentations” Generates
fake images with labels...

Scaling, Rotating, Shearing,
Elastic Transformation, Grid
Distortion, Gaussian Noise,
Motion Blur, Hue, Saturation,
Brightness, Contrast.



THE AI MODEL

YOLOv11

INSTANCE SEGMENTATION

Training set of over **1000 images** with **12 classes** used to train the AI model that does: detection, quantification and segmentation.

This is a pre-trained model, YOLOv11m-seg has 253 neural layers and 22,368,468 parameters.

Training settings: 25 epochs, image size **1280 pixels**, batch size of 16.

Using GPU chip: Nvidia RTX 4070 super... **9 hours**

Using CPU chip... **150 hours**

YOLOv11

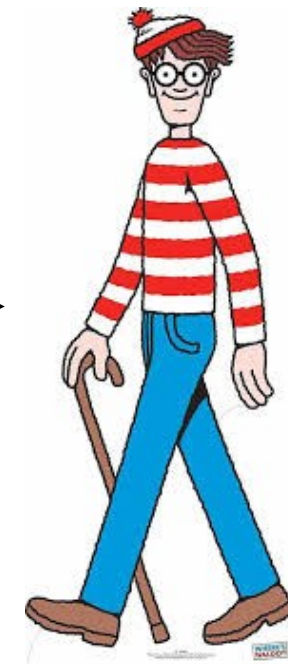
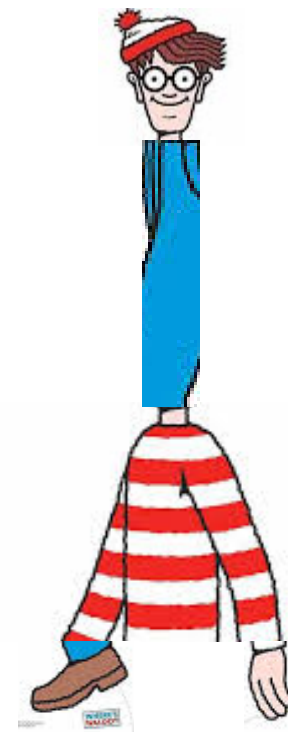
- ANALOGY -

Where's Wally?

You Only Look Once



Features



Feature Aggregation



Feature Maps

YOLOv11

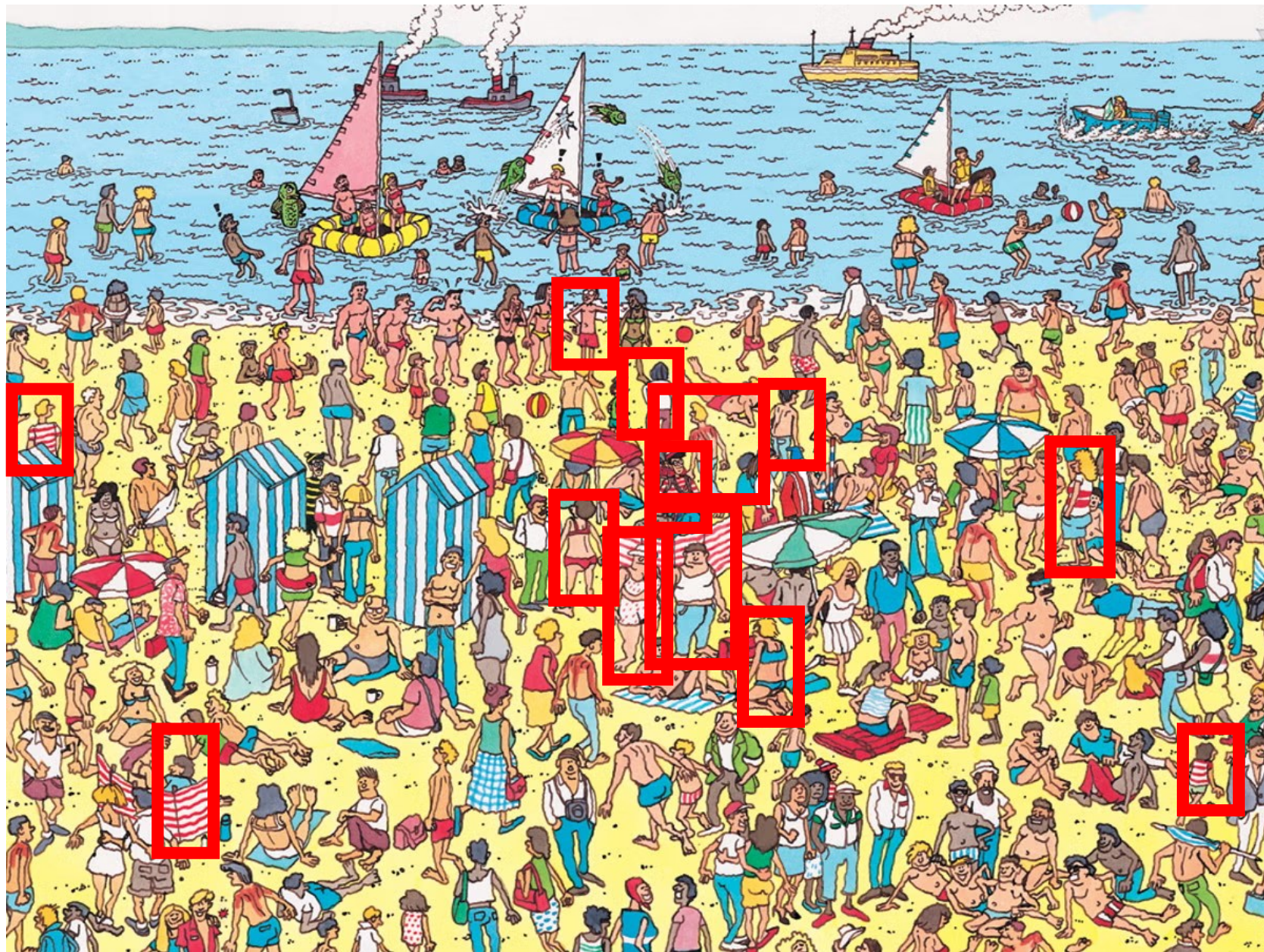
- ANALOGY -

Where's Wally?

You Only Look Once



Refined Feature Map



YOLOv11

- ANALOGY -

Where's Wally?

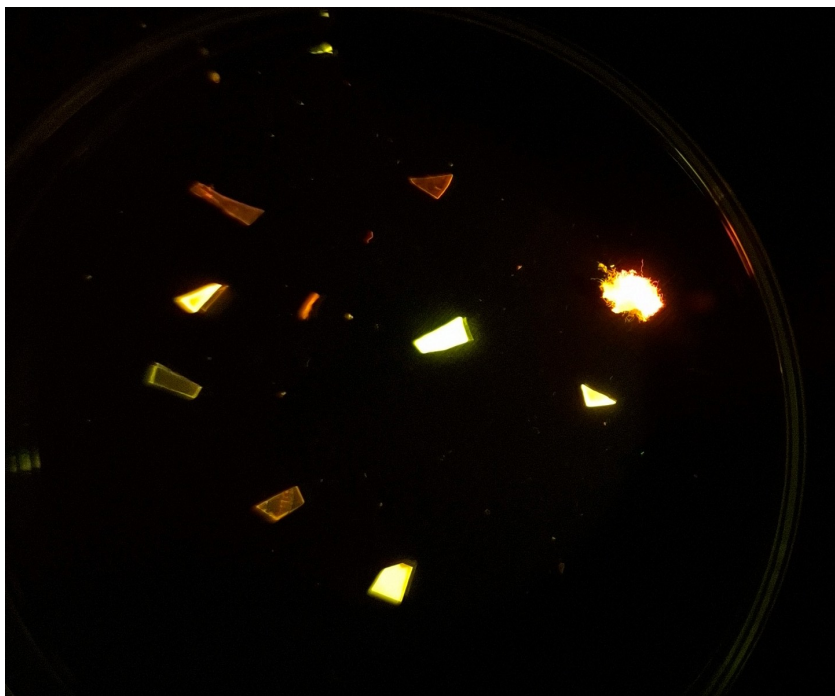
You Only Look Once



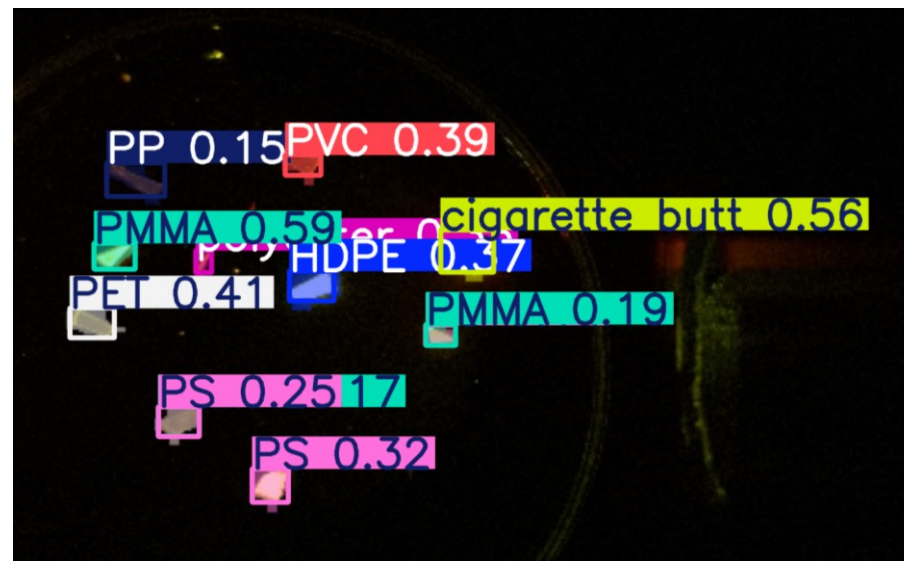
Refined Feature Map



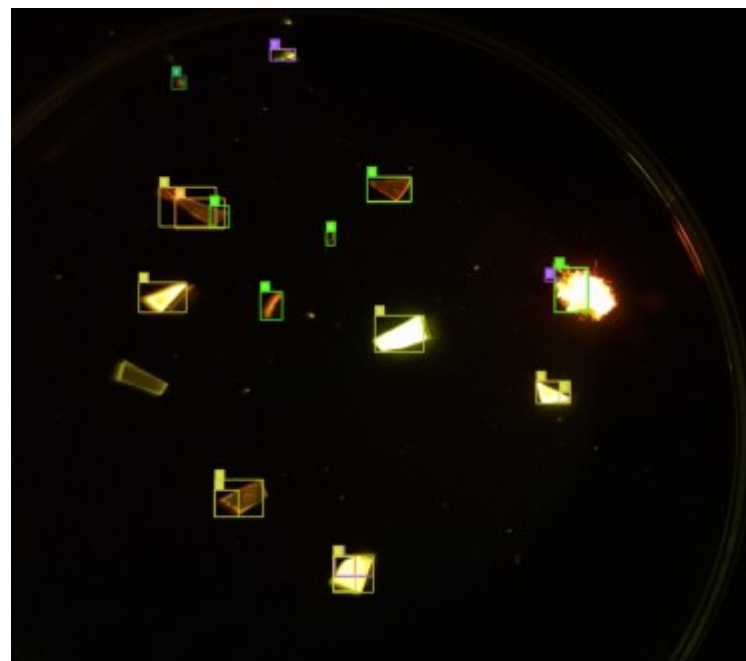
RESULTS



Original image



Output of image
without SAHI



Output of image
with SAHI

CONCLUSIONS

1

Understood & researched about YOLO

2

Created artificial plastic samples

3

Stained artificial plastic samples

4

Captured 405 samples pictures

5

Labelled 405 samples pictures

6

Understood & researched Data augmentation "Albumentations"

7

Created >1000 fake pictures & labels

8

Developed & Tested custom YOLOv11

9

Understood & researched about SAHI

10

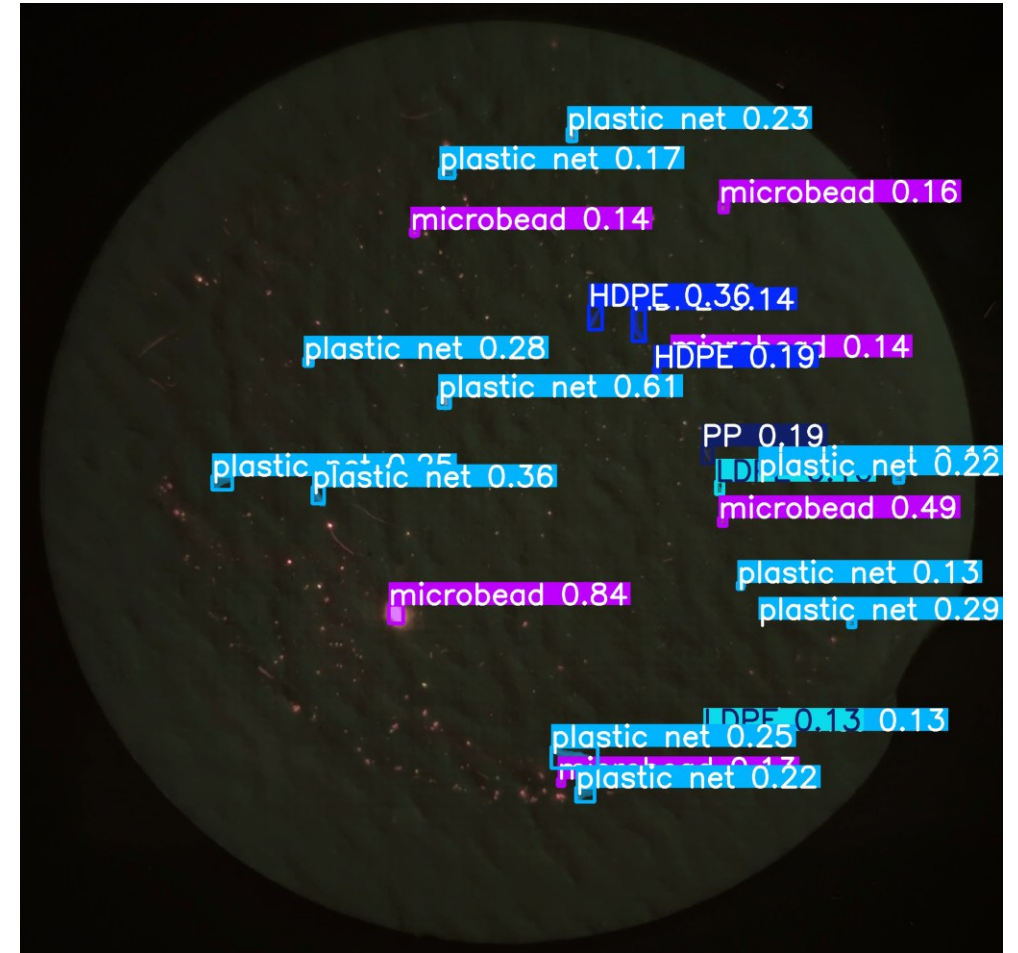
Developed the custom YOLOv11 + SAHI and wrote SOP to use

Thank you!

APPENDIX – Results on Real Samples



640x640 pixel version

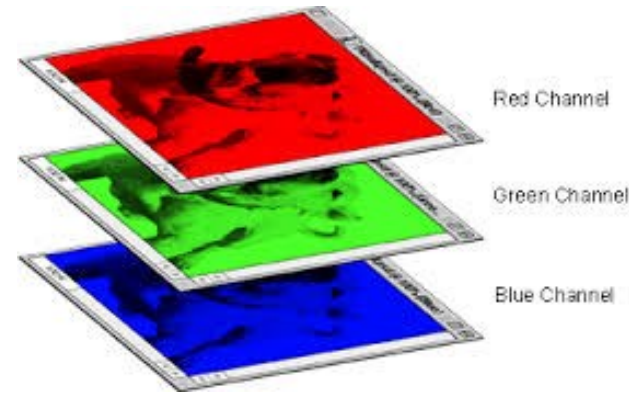


1280x1280 pixel version

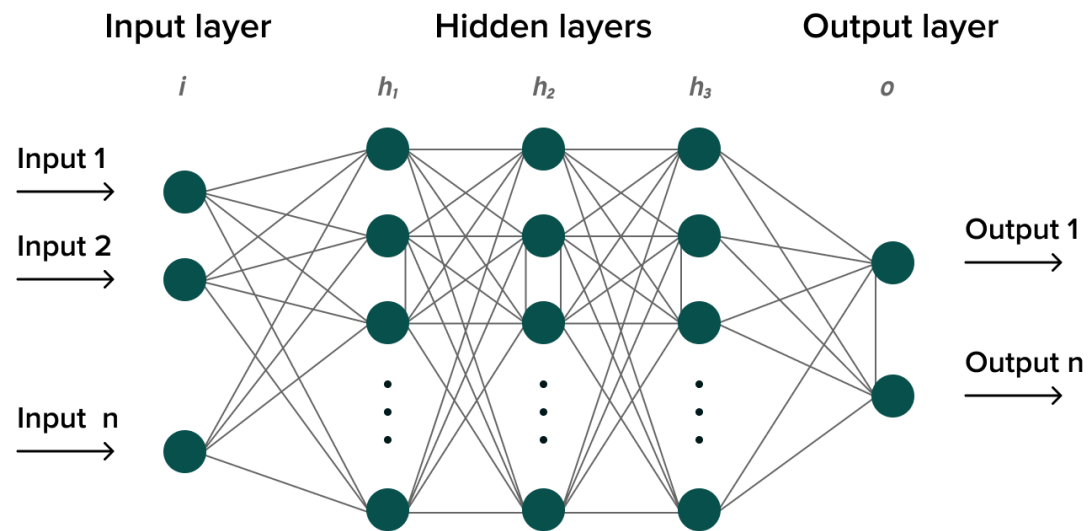
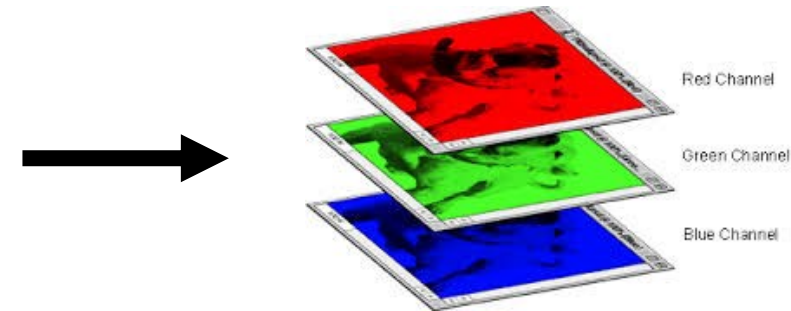
YOLOv11

BACKBONE

3000 x 4000 pixel
smartphone image

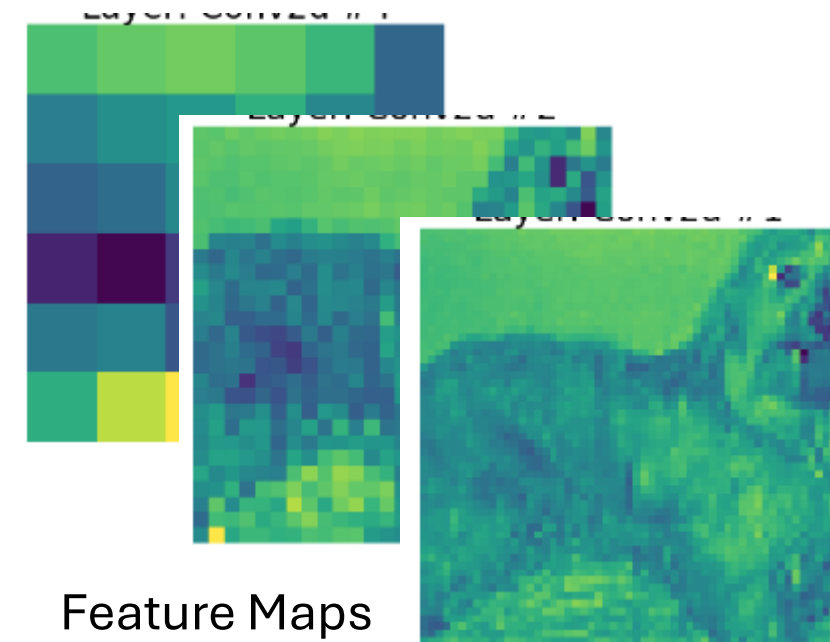


1280 x 1280 pixel
resize



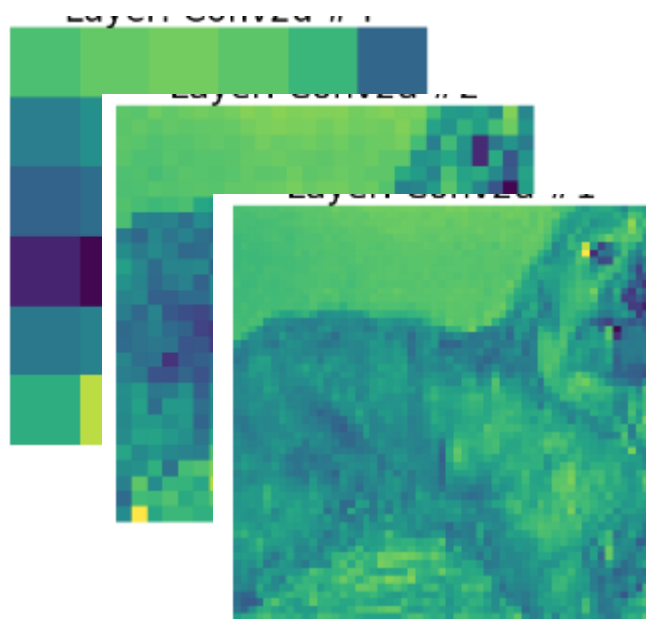
Convolutional Neural Network

Spatial pyramid
pooling fast
(SPPF)



YOLOv11

NECK

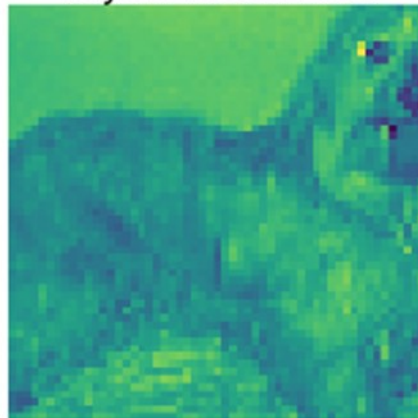


Feature Maps of
various resolutions

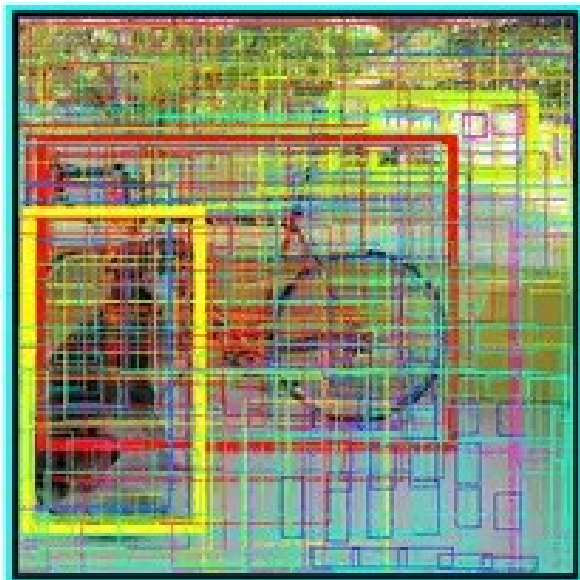


YOLOv11

HEAD



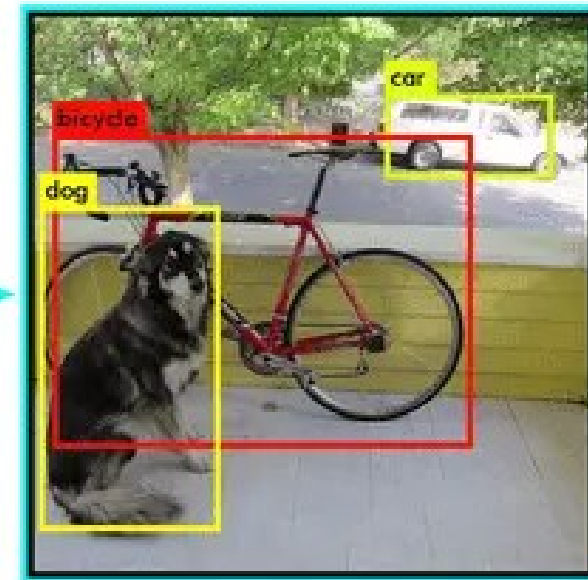
Based on refined feature maps, input image is predicted.



Box Generation
(Anchor free methods)

Non maximum suppression
(based on confidence scores)

IoU - Box overlapping area

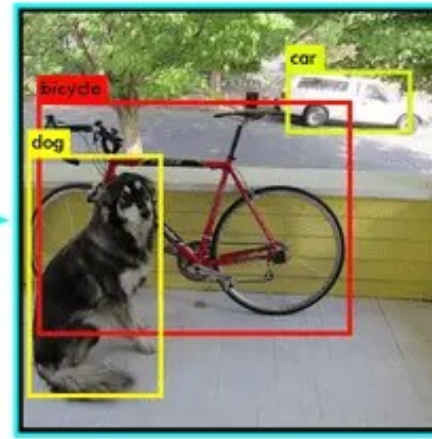


Final prediction
(Location, Class, Confidence)

YOLOv11

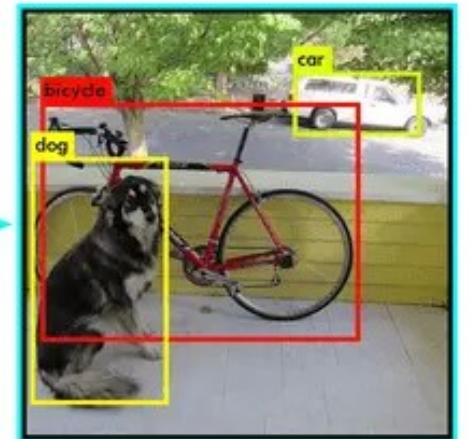
TRAINING STAGE

FINAL PREDICTION is compared with **GROUND TRUTH** training image. The difference between the location of the boxes, the classes are calculated via **LOSS FUNCTIONS**: Stochastic Gradient Descent, Adaptive Moment Estimation. Aims to reach the lowest non zero value through training and weight adjustments.



Final prediction

Compare



Ground truth

LOSS FUNCTIONS are used to adjust **weights** which optimised the YOLO model **performance**.